

ROI ET, Thailand

WMO: 484050

Lat: 16.052N

Lon: 103.664E

Elev: 467

StdP: 14.45

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	58.1	60.6	47.4	49.3	66.2	50.1	54.6	67.0	13.4	76.0	12.0	77.0	2.1	30	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.8	100.1	77.2	98.1	77.2	96.2	77.4	81.5	90.3	80.9	89.7	80.3	89.1	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.6	156.4	84.3	79.0	153.0	83.9	78.3	149.7	83.7	45.6	90.3	45.0	90.1	44.4	89.2	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.7	8.5	DB	53.7	102.7	2.8	1.8	51.7	104.0	50.1	105.0	48.5	106.0	46.5	107.3	
			WB	49.5	83.3	3.1	1.6	47.3	84.5	45.5	85.5	43.8	86.4	41.6	87.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	75.4	79.3	83.8	86.6	85.8	85.2	83.9	83.2	82.5	81.4	78.8	75.2
	DBStd	5.28	4.72	5.54	5.23	4.12	3.09	2.52	2.62	2.25	2.16	2.59	3.85	4.78
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	4	1	1	0	0	0	0	0	0	0	0	0	2
	CDD50	11595	788	820	1047	1098	1110	1056	1050	1029	976	974	864	783
	CDD65	6123	324	401	582	648	645	606	585	564	526	509	414	319
	CDH74	69129	2959	4313	7156	8493	8069	7496	6776	6357	5645	5226	3867	2772
CDH80	30734	1150	2044	3793	4686	4009	3463	2784	2406	1965	1932	1504	998	
Wind	WSAvg	2.5	2.5	2.2	2.3	2.2	2.2	3.0	3.2	2.9	1.7	2.1	2.7	3.0
Precipitation	PrecAvg	57.5	0.2	0.7	1.3	3.5	8.2	7.8	8.6	10.7	12.1	3.9	0.6	0.1
	PrecMax	73.5	1.6	3.9	5.3	7.1	19.0	11.7	16.0	18.0	22.9	8.8	2.5	1.0
	PrecMin	41.1	0.0	0.0	0.0	0.4	2.2	3.5	3.8	1.9	4.8	0.7	0.0	0.0
	PrecStd	8.8	0.4	1.1	1.4	2.0	4.4	2.4	3.3	4.8	4.8	2.3	0.8	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	98.0	101.6	103.7	100.8	97.4	96.1	93.6	92.7	92.6	94.1	92.7
		MCWB	73.0	74.1	75.0	77.9	79.5	79.3	79.3	79.8	79.9	78.3	77.4	74.6
	2%	DB	91.2	95.7	99.3	100.8	98.1	95.2	93.5	91.8	90.9	90.9	91.7	90.4
		MCWB	72.7	74.4	75.2	77.7	79.3	79.1	79.1	79.3	79.3	77.7	76.6	74.1
	5%	DB	89.1	93.5	97.2	98.5	95.7	93.4	91.5	90.4	89.6	89.6	89.8	88.2
		MCWB	72.3	74.2	75.7	78.0	79.2	79.2	78.9	79.1	78.9	77.2	75.6	73.0
	10%	DB	86.6	90.9	94.8	96.1	93.4	91.8	89.9	88.9	88.1	88.3	87.8	85.9
		MCWB	71.3	73.9	75.8	77.9	78.8	79.0	78.7	78.8	78.4	76.8	74.5	71.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	78.9	79.7	81.3	82.1	82.0	81.4	81.7	81.7	80.5	80.0	77.3
		MCDB	87.7	92.1	92.3	94.7	93.1	91.1	89.4	88.5	88.0	87.8	89.3	87.9
	2%	WB	75.6	77.4	78.6	80.1	81.0	81.0	80.6	80.9	80.8	79.7	78.7	75.7
		MCDB	86.4	90.1	92.3	94.0	92.0	90.1	88.6	87.9	87.3	86.7	88.0	86.5
	5%	WB	74.1	76.3	77.8	79.4	80.4	80.4	79.9	80.1	80.1	79.0	77.5	74.6
		MCDB	84.9	88.5	91.4	93.0	91.2	89.3	87.8	87.1	86.4	85.8	85.9	85.3
	10%	WB	72.5	75.2	77.0	78.8	79.8	79.8	79.3	79.6	79.6	78.3	76.4	73.2
		MCDB	83.8	86.6	90.5	92.2	90.0	88.4	87.0	86.4	85.5	84.9	84.4	83.7
Mean Daily Temperature Range	5% DB	MDBR	21.7	21.6	19.8	17.8	14.8	12.8	11.6	11.0	10.9	13.3	16.9	19.6
		MCDBR	22.8	22.7	21.7	20.1	17.5	14.9	13.8	13.0	12.8	14.7	18.3	20.6
	5% WB	MCWBR	7.6	6.5	5.5	4.7	4.2	3.8	3.9	4.1	4.2	4.7	5.9	6.9
		MCWBR	19.8	19.6	19.0	18.2	15.5	13.4	12.5	12.0	11.8	12.8	15.9	18.9
Clear-Sky Solar Irradiance	taub		0.477	0.595	0.664	0.620	0.514	0.470	0.460	0.464	0.507	0.525	0.471	0.465
	taud		2.084	1.788	1.678	1.815	2.137	2.279	2.308	2.287	2.155	2.073	2.197	2.147
	Ebn at Noon		256	231	220	231	253	261	265	266	256	247	257	256
	Edh at Noon		49	70	80	70	50	43	42	43	49	52	44	46
All-Sky Solar Radiation	RadAvg		1568	1653	1680	1809	1767	1654	1551	1508	1474	1600	1569	1514
	RadStd		87	81	72	79	95	81	114	99	110	108	56	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	+0.45	N/A	N/A	N/A	+0.77	+0.67	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	+0.49	N/A	N/A	N/A	+0.27	+0.23	N/A	N/A	N/A	N/A

Nomenclature: See separate page